

AI-Powered Sculpting: Reviving Digital Sculpture with Generative Workflows

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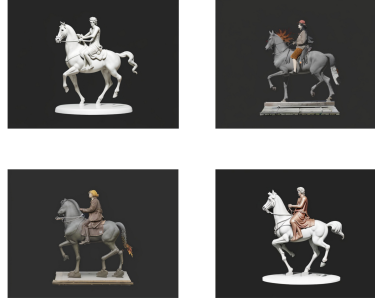
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a. Basemesh



b. AI Interpretations



c. Composite



Figure 1: Process of Using AI Diffusion Models to Ideate Through Basemesh Transformation

Abstract

This paper presents a minimalist hybrid workflow that couples digital sculpting with generative image synthesis for character look development. A manually sculpted bust in ZBrush serves as the authoritative base; camera-matched renders are processed with Stable Diffusion (image-to-image and inpainting) to propose stylistic variations and local ornament while preserving volumetric intent. Treating the model as an instrument beside craft accelerates exploration without displacing authorship. Qualitative studies show that prompt discipline and conservative denoise ranges yield coherent, review-ready outcomes; approved decisions are round-tripped back into geometry so they persist beyond 2D imagery. The approach demonstrates a pragmatic, reproducible pathway for co-creative sculpt workflows leveraging latent diffusion and established digital sculpt practice.

CCS Concepts

• Computing methodologies → Geometry and Modeling.

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1 Introduction

Stable Diffusion models have broadened visual ideation, achieving state-of-the-art image synthesis and enabling controllable conditional generation; yet unconstrained sampling can erode anatomical fidelity and plane hierarchy [Dhariwal and Nichol 2021; Ho et al. 2020]. We therefore position Stable Diffusion as a complementary instrument: ZBrush establishes proportion; Stable Diffusion proposes material cues and ornament from fixed camera renders. By keeping the sculpt authoritative and feeding accepted motifs back into geometry, the workflow preserves form-centred authorship while exploiting generative breadth within a compact toolset [Ho et al. 2020].

2 Related work

Stable Diffusion operationalises latent diffusion—denoising in a compressed latent space—to deliver high-resolution synthesis with practical compute and strong prompt-image alignment [Rombach et al. 2022]. Editing and translation modes relevant to our pipeline include img2img-style conditioning and inpainting with diffusion priors, which have been studied in general frameworks such as Palette (image-to-image diffusion), SDEdit (editing via stochastic differential trajectories), and RePaint (mask-based inpainting) [Saharia et al. 2022]. ZBrush remains the de facto environment for blocking anatomy and editing planes with fine local control, anchoring character workflows in durable geometry rather than fragile paint-overs [Maxon and Pixologic [n. d.]]. Our contribution integrates these strands: Stable Diffusion is used for stylistic readings and localised inpainting, but the sculpt is the source of truth, and accepted decisions are re-established as geometry [Maxon and Pixologic [n. d.]; Saharia et al. 2022].

3 Workflows

Previous digital sculpture toolchains have prioritised either full 2D paint-over or heavyweight texturing stacks; in this work, our goal is to accelerate character look development while keeping geometry authoritative. We identified three practical workflows (see Figure 2) that bring a sculptor’s intent to screen efficiently using only ZBrush and Stable Diffusion.

ZBrush basemesh. All our workflows begin with a basemesh sculpted in ZBrush, blocked for proportion, silhouette, and primary anatomical planes. The goal at this stage is not polish but stability: by establishing a neutral pose with clean edge flow and controlled form transitions, we provide a reliable anchor for subsequent AI reinterpretations. Consistent camera placement and grey-clay renders ensure reproducibility, while segmentation passes (Flat/ID masks) allow later selective inpainting without leaving the sculpting environment. Unlike traditional pipelines that move rapidly to UVs and textures, we prioritise geometry as the authoritative representation of design intent, postponing surface detail until AI-guided exploration identifies promising motifs.

AI interpretations. From each basemesh render, we generate multiple stylistic variations using Stable Diffusion [Rombach et al. 2022] and SDXL [Podell et al. 2023] in img2img mode. Rather than training character-specific models, we rely on disciplined prompting, low denoise values (0.35–0.55), and seed reuse to preserve volumetric intent while exploring alternative materials, costumes, or ornamental vocabularies. These interpretations act as speculative design probes: hairline trims, saddle details, or bronze patina treatments appear with minimal sculptor overhead. To control drift, we adopt noun-first prompt templates and reuse identical camera renders, a strategy aligned with prior findings on diffusion stability under constrained schedules [Ho et al. 2020]. The resulting outputs are curated into motif boards, where accepted silhouettes or surface features are fed back into ZBrush via alphas or stencils, thereby translating ephemeral AI proposals into durable 3D geometry.

Composite renderings. For presentation and evaluation, AI interpretations are composited into contextual scenes—ranging from painterly backdrops to photo-real environments. This stage is crucial for judging plausibility and narrative fit: a sculpture that reads as coherent in isolation may behave differently when lit, staged, or juxtaposed against real-world cues. Using conservative inpainting masks, we integrate AI-generated riders and horses into landscapes or gallery settings, producing visual hypotheses for how the piece might exist in situ. These composites serve two functions: they expand stylistic search by situating the work in diverse cultural idioms, and they provide communication-ready frames for discussion with collaborators. This approach is consistent with recent work showing the effectiveness of locality-preserving diffusion inpainting [Lugmayr et al. 2022]. Across this three-stage workflow we observed that prompt libraries (structured as material + craft verb + style idiom), consistent camera-locking, and deliberate seed reuse yield the most stable and legible results. By anchoring geometry first and iterating stylistic search through AI, we accelerate look development while retaining sculptural authorship. This approach contrasts with conventional pipelines that often privilege surface

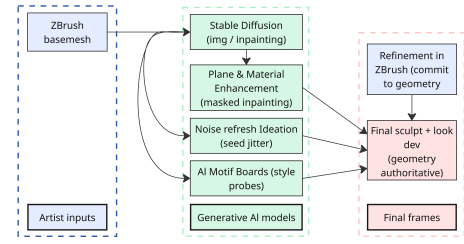


Figure 2: Workflow for AI-assisted digital sculpting.

texturing or 2D paint-over; instead, our method treats AI as a collaborator in motif discovery while keeping ZBrush central to the decision-making process [Ho et al. 2020; Rombach et al. 2022].

4 Conclusion

AI-assisted workflows allowed sculptors to devote more time to form, silhouette and material storytelling, and less to tedious paint-over or exploratory texturing passes. However, substantive manual clean-up in ZBrush was required to meet our quality bar, and over the course of production we repeatedly refined our prompt templates, seed discipline and denoise windows to build intuition. The inpainting workflow demanded unusually clean inputs—neutral clay renders, stable lighting and crisp masks—cleaner than many quick look-dev tests we would typically accept. Nonetheless, latent-diffusion proposals exceeded our initial expectations, enabling a compact team to complete numerous coherent concept studies in weeks rather than months while keeping the sculpt authoritative. We expect results to improve as base checkpoints incorporate more sculptural and ornamental exemplars, and as lightweight conditioning strategies further constrain anatomy, plane hierarchy and palette without resorting to heavy fine-tuning. Additional qualitative studies and ablations (seed reuse, denoise ranges, camera consistency) are included in the supplementary materials to facilitate replication.

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